

The 10th International Conference on Computer Science and Computational Intelligence 2025

Aqualyze: Smart Monitoring Water Quality through IoT-Enabled Sensors in RAS

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Abstract

Mangrove crab (*Scylla serrata*) cultivation faces challenges in maintaining optimal water quality, which is crucial for their health and growth. Traditional water quality monitoring methods can be labor-intensive and provide only intermittent data. This research presents "Aqualyze," an Internet of Things (IoT)-based system designed for smart, real-time monitoring of water quality in Recirculating Aquaculture Systems (RAS) for mangrove crab cultivation. The Aqualyze system employs sensors to continuously measure key parameters: water temperature (DS18B20), pH (4502C), dissolved oxygen (DO) (SEN0237), and turbidity (TS-300B). Data gathered by these sensors is processed by an ATmega328P microcontroller integrated with an ESP8266 Wi-Fi module. The processed data is then transmitted via the MQTT protocol to a HiveMQ broker and subsequently stored in a Firestore database for logging and analysis. Real-time monitoring results are displayed on an LCD I2C and can be accessed by users via a LINE bot, which provides alerts and on-demand data. The result of testing showed that while pH (7.1–8.1) and temperature (28.50–28.80 °C) remained within acceptable ranges, dissolved oxygen was consistently low (2.1–3.0 ppm), and turbidity fluctuated significantly with occasional sensor errors due to improper placement. This system aims to provide aquaculture operators with timely and accurate water quality information, enabling proactive management and potentially improving crab survival and yield.

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Peer-review under responsibility of the scientific committee of the 10th International Conference on Computer Science and Computational Intelligence (ICCSCI) 2025

Keywords: Mangrove Crab; Internet of Things; water quality monitoring; aquaculture; RAS

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1. Introduction

Indonesia is the largest archipelagic country in the world, with a total water area of 6.4 million km², or two-thirds of the total area of Indonesia [1]. Currently, Indonesia is also one of the largest crab exporters in the world. One of the types of crab that is often cultivated is mangrove crab, or mud crab (*Scylla serrata*). This type of crab lives in mangrove areas and in the sea. This crab is high in protein, low in fat, and high in unsaturated fatty acids. Crabs are omnivorous and cannibalistic animals and are active at night (nocturnal).

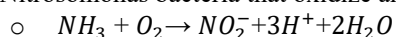


Fig. 1. *Scylla Serrata*.

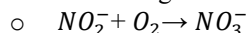
However, the process of crab cultivation is not easy; there are several factors that can cause death in crabs. So, it's necessary to maintain good water quality for crab cultivation. The following are important factors that need to be considered in crab cultivation:

- Water quality includes several things, one of which is salinity, which plays an important role in the life of crabs. An optimal salinity level environment for crabs is around 15-31 ppt [2].
- pH water, the acidity level of water; the lower the pH, the more acidic the water. When the pH of the water is high, then the water is more alkaline. pH is one of the determining factors for crabs to be able to change their skin (molting). The appropriate pH value for crabs is in the range of 6.5–7.5 pH [2].
- The water temperature of crabs usually lives in mangrove or coastal areas where the appropriate temperature for crab cultivation is between around 26–31 degrees Celsius [2].
- Dissolved oxygen is very necessary to increase oxygen diffusion in water to be higher. For a good enough oxygen level, it must be above > 5 ppm [3].
- Ammonia is dirt or leftover food from crabs, which becomes bacteria and then decomposes into nitrate and nitrite. In this prevention, RAS (Recirculating Aquaculture System) can be used [4]. A good RAS system can reduce nitrification, which is the formation of nitrite (NO₂) or nitrate (NO₃) from ammonia (NH₃) [5]. This process occurs due to 2 stages:

By Nitrosomonas bacteria that oxidize ammonia or ammonium into nitrites [6].



By bacteria of the genus Nitrobacter that oxidize nitrites to nitrates [6].



In this study, crab cultivation will be carried out in crab houses that have applied the RAS system. From the result of factors to maintain water quality, it is generally done by conventional methods, such as temperature measured using a temperature meter, pH with a pH meter, and oxygen level with a dissolved oxygen meter. So, it requires many tools that must be prepared to determine the quality of water for crab cultivation. With the current application of IoT (Internet of Things) technology, this research will make a measurement scheme for water quality in a conventional way into a technology that is integrated with the internet or known as IoT [7]. However, this research has a limitation in that measurements will not be made for ammonia; this requires lab tests to be able to produce accurate results. The measurement results by using IoT technology will create a system called Aqualyze by producing measurement results that consist of temperature, pH, dissolved oxygen, and turbidity [8] to show in the Instant Messaging Application [9].

2. Research Methodology

The IoT-based water quality monitoring system developed in this study utilizes four primary sensors, including the 4502C pH sensor, SEN0237 dissolved oxygen (DO) sensor, DS18B20 temperature sensor, and TS-300B turbidity sensor. An Atmega3284P with ESP8266 served as a microcontroller to collect data from sensors [10]. Measurement results are presented on an LCD I2C and transmitted to a monitoring platform via MQTT protocol. Firebase collects the data from MQTT to store in the Firestore database. Users can receive monitoring data via a LINE bot.

2.1. Research Process

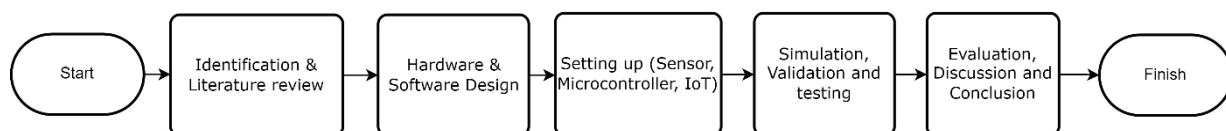


Fig. 2. Research Flowchart.

Based on Figure 2, the first stage is identification and literature review. At this stage, it is obtained from various sources of literature, including journals, books, or websites. This identification and literature review process is a preparatory stage for researchers in understanding the parameters and architecture of IoT that can be used in aquaculture. Hardware and software design is the process of preparing materials and assembling devices, which has the following arrangement: Sensors → Microcontroller (ESP32) → Wi-Fi → Cloud or Direct API → LINE Notify. By preparing the components in table 1 as follows:

Table 1. Component

Component	Device	Function
Microcontroller	ATmega328P + ESP8266	Sensor reading, processing, and sending data
Temperature	DS18B20	Measure temperature
pH	4502C	Monitor acidity
DO	DFRobot SEN0237	Measure dissolved oxygen
Turbidity	TS-300B	Check clarity
Power	Power Supply 12V + Stepdown DC-DC 5V	Continuous operation

At the next stage after all components are assembled, validation is carried out; in this case, it is calibrated for each sensor using standard solutions. After getting the results, an evaluation is carried out to get reliable solution results.

2.2. Internet of Things (IoT)

The Internet of Things (IoT) refers to a network of interconnected physical objects, or "things," equipped with sensors, software, and connectivity, enabling them to collect and exchange data over the internet. A typical IoT system consists of sensors that acquire data from the physical environment, microcontrollers that process this data, network connectivity (wired or wireless) for data transmission, cloud-based platforms for data storage and analysis, and user applications allowing data visualization and interaction [11]. IoT technology has widespread applications across various sectors, including industrial automation, smart homes, environmental monitoring, and healthcare. In aquaculture, IoT is increasingly utilized for various purposes, such as optimizing feeding schedules, detecting diseases early, monitoring water quality in real time, and employing predictive analytics to improve farm

management [12]. The ability of IoT to facilitate continuous data collection and remote monitoring of crucial water quality parameters makes it particularly relevant for aquaculture systems like RAS, where maintaining a stable and optimal environment is key to successful mud crab farming. The core value proposition of IoT in this context lies in providing aquaculture farmers with timely and accurate information, enabling them to make informed decisions and react promptly to any changes in water conditions, ultimately leading to enhanced efficiency and reduced operational risks. Furthermore, the decreasing costs and increasing accessibility of IoT components make these advanced monitoring solutions more feasible for a broader range of aquaculture operations, including smaller-scale crab farms. The proposed Aqualyze monitoring system can connect through internet service, and it can be able to communicate with another network device.

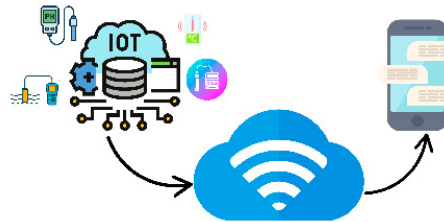


Fig. 3. The proposed structure of the IoT monitoring system.

Figure 3 presents the proposed structure of the embedded monitoring system. The Analyze system monitoring can be a type of microcontroller, which contains an integrated circuit. The Arduino Uno R3 is easy to use and can connect to Wi-Fi [13]. This integrated circuit relates to some sensors, which are temperature, pH, DO, and turbidity, to monitor water quality. This design also can send data using API and send the request data to instant messaging.

2.3. Recirculating Aquaculture System (RAS)

Recirculating Aquaculture Systems (RAS) have emerged as a significant technological advancement in mud crab aquaculture, offering a controlled environment where water conditions can be stabilized and optimized [14]. A typical RAS comprises several key components, including culture tanks for housing the crabs, mechanical filtration to remove particulate matter, biological filters to convert harmful ammonia into less toxic nitrates, protein skimmers to eliminate dissolved organic compounds, and ultraviolet (UV) sterilizers to disinfect the water. The implementation of RAS offers numerous benefits, including a substantial reduction in water usage, minimized environmental impact due to reduced wastewater discharge, enhanced biosecurity, and the possibility of establishing crab farms in locations without direct access to brackish water sources [15]. In contrast, traditional earthen pond systems often face challenges such as water pollution from external sources like agricultural runoff, inconsistencies in seawater quality, and difficulties in precisely regulating salinity levels, which can negatively affect mud crab production. The increasing adoption of RAS in mud crab farming highlights the industry's move towards more controlled and intensive cultivation methods, where maintaining stable and optimal water quality is paramount for maximizing productivity. Figure 4 is an RAS design that will be implemented.

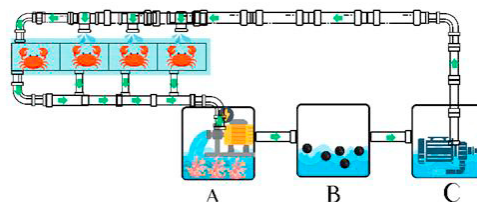


Fig. 4. RAS design on crab house.

According to the design of RAS, here are some detailed explanations for each tub.

- A. Mechanical filters mainly filter solid impurities such as sand, food scraps, crab droppings, or other particles. The tools used are Dakron, skimmer protein, and oyster mussels.
- B. Biological filters decompose the first filtered droppings into ammonia substances derived from food waste and crab droppings. The system of IoT will be implemented in this box.
- C. The chemical filter mainly functions as a biological disinfectant. This process also plays a role in increasing the chemical compounds needed for crab cultivation. The tool used in this chemical filter is a UV lamp placed on that tub.

2.4. Instant Messaging Application

Integrating Internet of Things (IoT) technologies in aquaculture has enabled efficient real-time monitoring and automation in crab farming. Pitakphongmetha et al. proposed an innovative crab farming system utilizing sensors for water quality, automatic feeding, and image-based molting detection, significantly enhancing survival rates and farm productivity through timely alerts and data visualization via platforms like LINE [16]. Boonsong's study reinforces the critical importance of dissolved oxygen (DO) monitoring, using an embedded wireless system with NodeMCU-ESP8266 to capture the inverse relationship between water temperature and DO, thereby ensuring optimal aquatic conditions [9]. Meanwhile, Steinberg's analysis of LINE as a super app highlights its role beyond messaging, enabling automation through services like content delivery, sticker markets, and direct-to-user notifications, making it a strategic interface for alerting and engaging crab farmers through personalized updates and automated reports [17].

2.5. Hardware and Sensor Setup

The device consists of a 4502C pH sensor, a DS18B20 temperature sensor, a SEN0237 dissolved oxygen sensor, a TS-300B turbidity sensor, an ATmega328P microcontroller with ESP8266, a 16x16 I2C LCD, a DC-DC step-down DFR0379, and a 12V power supply. The ATmega328P and ESP8266 are programmed separately through the Arduino IDE. Pin Po on the 4502C pH sensor is connected to pin A0 on the ATmega328P. The blue data wire on the DS18B20 temperature sensor is connected to pin A1 on the ATmega328P [18]. The blue wire of the dissolved oxygen sensor SEN0237 is connected to pin A2 of the ATmega328P. Pin A of the turbidity sensor is connected to pin A3 of the ATmega328P. The sensor measurement results are displayed on the I2C LCD by connecting the SCL and SDA pins of the I2C LCD to the ATmega328P. The power of this circuit is obtained through a 12V power supply converted to 5V through a DC-DC DFR0379 step-down. All GND and VCC pins from the sensor and Arduino are connected to the GND and VCC pins of the step-down DC-DC DFR0379. The GND pin on the ATmega328P is also connected to the sensor GND pin circuit. Especially for the DS18B20 temperature sensor, the VCC cable is first connected to a 4.7k ohm resistor to adjust the power supplied to the sensor. The device is arranged as shown in Figure 5.

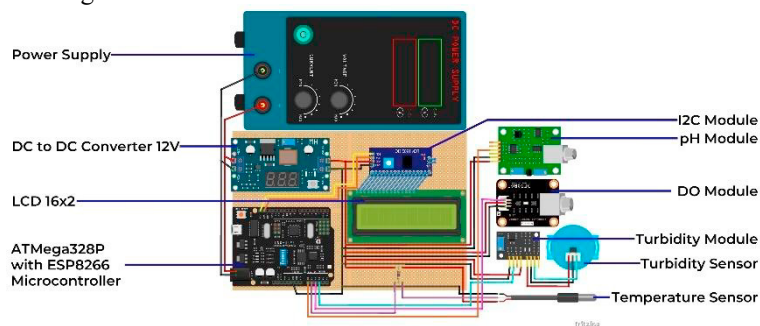


Fig. 5. Fritzing electronic circuit.

2.6. Sensor Calibration

The sensor devices were calibrated to ensure the data readings gave valid results. The pH sensor is calibrated using reference voltage data at pH 4, pH 7, and pH 9. The known reference voltages are at least two of the three options. A special pH powder dissolved in a 250 mL distilled water solution is required for reference voltages at all three pHs. The reading of the sensor voltage in this solution will be the main parameter for the voltage-to-pH conversion calculation formula in the Arduino program. Dissolved oxygen sensors are calibrated using single-point method. The sensor probe is exposed to the ambient air while the voltage value is recorded. This value will be used as the reference point for subsequent measurements. Regular calibrations are required to ensure the sensor's accuracy and reliability. Temperature and turbidity sensors do not require calibration.

2.7. Communication and Data Flow

The ATmega328P microcontroller and ESP8266 are arranged on the same board, but a serial communication line is required to connect the data. The program for the ATmega328P will collect voltage readings from the sensor to be converted into appropriate units and compiled into JSON format for later printing to the serial. The program for ESP8266 is set to read the serial message sent by ATmega328P and upload it to a topic in the MQTT broker by utilizing the Wi-Fi connection [19]. From the MQTT broker, the data cannot be directly connected to the Firestore database, so a Node.js program is needed. This program reads the latest available data from the MQTT broker, adds a timestamp, and then uploads it to Firestore. In Firestore, the data is stored in a collection and stored in time order to facilitate monitoring. The collected data can be accessed directly through the LINE bot Aqualyze, which is programmed to send data every noon and every time the user sends the command “/monitor”. LINE monitoring is achieved with settings in the LINE Developer Console, a Node.js cron job program for sending routine data, and a Node.js program that handles commands from the webhook.

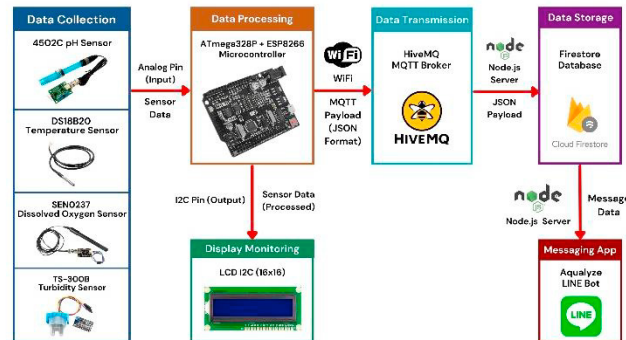


Fig. 6. Device data flow.

3. Results and Discussion

3.1. Device Realization

Device system realization can be seen in Fig. 7 below:

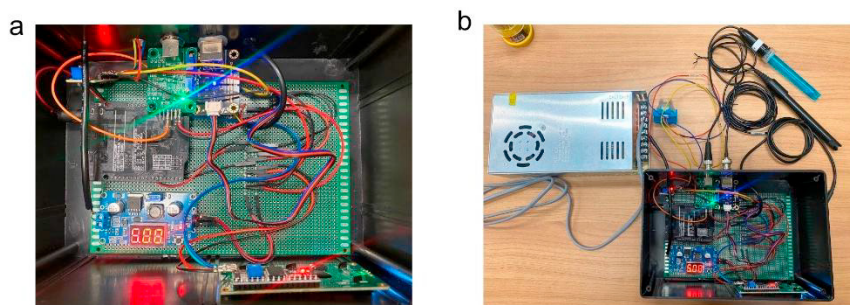


Fig. 7. (a) Device internal; (b) Device overhead view.

The entire circuit has been enclosed in a waterproof black ABS box to prevent water from entering the electrical components, possibly leading to short circuits or corrosion. All cables connected to the microcontroller and the DC-DC step-down converter are soldered directly to the PCB, reducing the risk of technical failures caused by broken connections between pins. On the back of the box, five openings have been created for various connections: the power supply line leading to the DC-DC step-down converter, the cable for the DS18B20 temperature sensor, the connection for the 4502C pH sensor module, the line for the SEN0237 dissolved oxygen sensor module, and the cable for the turbidity sensor module. Additionally, a cutout has been made on the front side of the box to accommodate the I2C LCD, allowing for direct monitoring of data from the sensors.

Water quality monitoring data was successfully sent through a LINE bot called Aqualyze. Users can send messages "/monitors" through the LINE application on mobile phones and desktops to facilitate access to monitoring anytime and anywhere. The data sent is accompanied by a class description, making it easier for users to know if a condition exceeds the optimal limit.

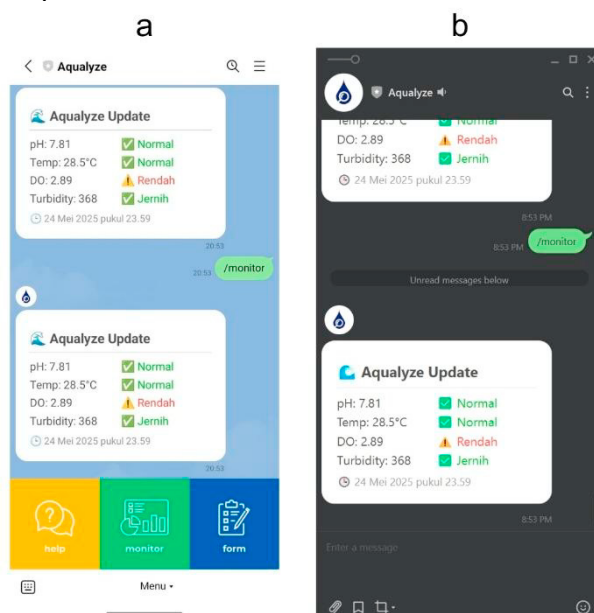


Fig. 8. (a) Line app mobile view; (b) Line desktop view.

3.2. Discussion

Data collection was conducted on a crab house in Semarang City on May 24, 2025, from 17:00 until 00:00 Western Indonesian Time with a data transmission interval of 2 seconds. Data was acquired from the Arduino sensor, then processed in ATmega328P with ESP8266, transmitted using HiveMQ MQTT, and finally stored in Firestore. The database was retrieved from the Firestore collection and downloaded in CSV format using Python code. It contains pH, temperature, dissolved oxygen (DO) values, turbidity, and timestamp. The time format from the timestamp in Firestore is UTC format, then converted to Western Indonesia Time (UTC+7). The data is in a time series graph, as shown in Fig. 9.

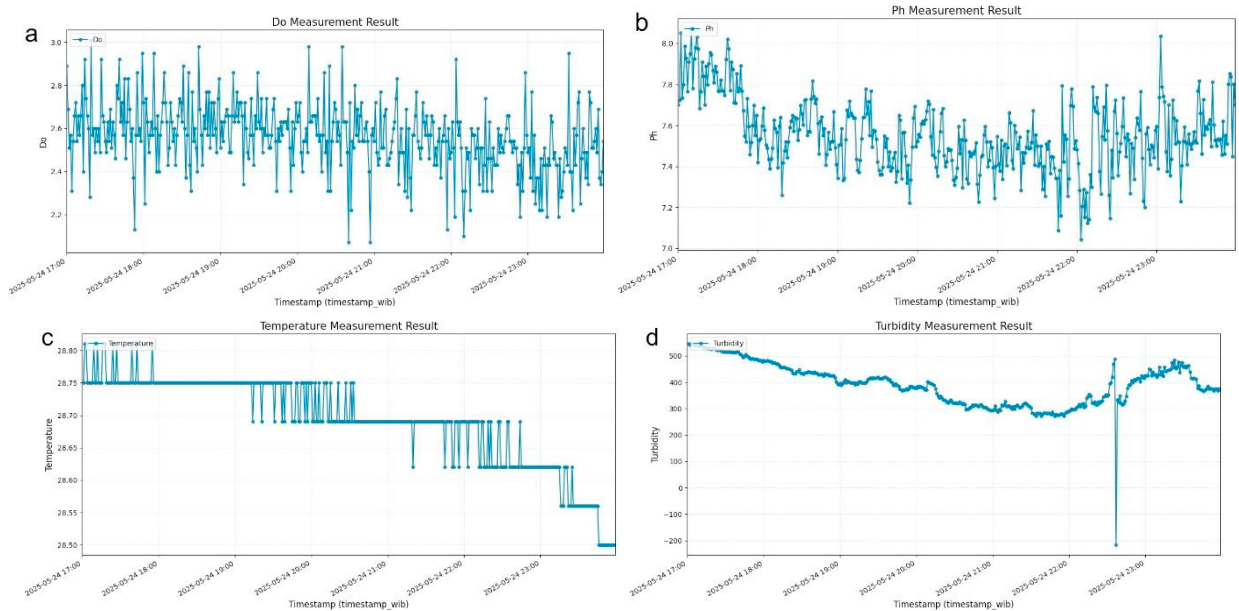


Fig. 9. (a) DO measurement results; (b) pH measurement results; (c) Temperature measurement results; (d) Turbidity measurement results.

The graph in Fig. 9 visualizes the measurement results from pH sensor 4502C, temperature sensor DS18B20, dissolved oxygen (DO) SEN0237 sensor, and turbidity TS-300B sensor. Figure 9a above shows that DO measurement results fluctuate from 3.0 to 2.1 ppm. The DO sensor is placed in the corner of the crab house, far from the water sprinkler. This decision was taken because if we put the sensor near the water sprinkler, it would fluctuate heavily because of the inconsistent oxygen in the flowing water. DO value of 3.0 to 2.1 ppm is categorized below standard since 5.0 ppm is suitable for mud crabs. Overall, the pH measurement values had a minimum value of 7.1 and a maximum of 8.1. A reasonable pH value that should be present in a crab house is above 7.5. For the most part, the measured pH values meet these standards, so the crab house can be classified as having a normal pH value. In detail, the pH value fluctuates from 8.1 to 7.7 from 17:00 to 17:30 and then decreases to 7.3 to 7.8 from 18:00 to 21:45. From 22:00 to 00:00, the pH fluctuation increases from 7.1 to 8.1.

The temperature data shows a measurement value that decreases over time. The crab house temperature ranged between 28.80 for the highest measurement at 17:00 and 28.50 at 00:00. This is due to the decreasing air temperature at night, which also causes the water temperature in the crab house to decrease. The air at night can affect the water temperature in the crab house because of the placement of the crab house in a semi-outdoor location. The temperature value of 28.80 to 28.50 degrees Celsius is still considered normal because the fair value limit is 26–31 degrees Celsius. Turbidity measurements showed a decrease from a value of about 545 NTU at 17:00 to 272

NTU at 21:30. Then, the turbidity value tends to rise again until it reaches the highest point of 500 NTU at 22:40 and around 23:30. At 22:37, the turbidity measurement showed an unusual value of -217. This unusual value situation could be caused by water entering the sensor, which interferes with the measurement data. The turbidity sensor must be placed right on the water's surface because water can enter the sensor and interfere with its electrical function if it is too deep.

4. Conclusions and Future Work

This research successfully developed and implemented "Aqualyze," an IoT-based system for real-time monitoring of temperature, pH, dissolved oxygen (DO), and turbidity in a mangrove crab Recirculating Aquaculture System (RAS). The system effectively collected data via sensors, processed it using an ATmega328P and ESP8266, transmitted it via MQTT, stored it in Firebase, and provided user updates through an LCD and a LINE bot. Testing on May 24, 2025, revealed that while pH and temperature were generally within acceptable ranges (7.1-8.1 pH; 28.50-28.80°C), DO levels (2.1-3.0 ppm) were below the recommended >5 ppm, and turbidity readings showed variability, including an anomaly attributed to sensor interference. Despite the acknowledged limitation of not measuring ammonia, Aqualyze demonstrated its value as a tool for improved aquaculture management by providing continuous, accessible water quality data, with potential for future enhancements like ammonia sensing and refined sensor protocols.

Author Contribution Statement

The study was conceived and designed by Violitta Yesmaya. The development system Aqualyze by Jeremy Sean, Justo, and Ignasius. With guidance in the development system together with Ms. Adhe Lingga Dewi and Mr. Dany Eka Saputra. The writing and review are guided by Suzana Ariff Azizan, Evawaty Tanuar, and Roikhanatun Nafi'ah. All authors read and approved the manuscript.

Aknowledgements

We acknowledge the support of the Bina Nusantara University International Research Fund (PIB) under Proposal No. PIBT-25057 titled "Monitoring Water Quality by Using IoT for Mangrove Crab (*Scylla Serrata*) in Recirculating Aquaculture System – RAS".

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